Can disease spread be modelled by a reaction-diffusion equation?

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GROUP 10

Abstract

In this study, we modelled the spread of disease in Manchester through a reaction-diffusion equation, comparing this to COVID-19 data in Boston. The main assumptions of the model were a constant population density, a recovery time of 14 days, and negligible effects on immunocompromised individuals. The Euler method was used to solve the equation computationally. Our model successfully recreated spatial disease spread when compared to COVID-19 data in Boston, although it diverged from the data after ∼30 days. The model could be improved by accounting for more advanced assumptions including a travelling population, multiple outbreak centres and behavioural changes.

Introduction

Modelling disease spread is crucial for governments to develop effective contingency plans for large-scale pandemics. The UK's "inadequate response" to COVID-19 serves as evidence for poor governmental action [1]. Motivated by the need for better predictive tools, we attempted to model disease spread within Greater Manchester using a reaction-diffusion equation, chosen because the random large-scale motion of people is analogous to Brownian diffusion.

However, due to mass conservation, a standard diffusion equation is incapable of accurately modelling disease spread, because diseases constantly generate new infections. A reaction term is therefore added to account for both new infections and patient recovery, yielding the SIR-based framework described below.

Theory - Equations and Assumptions

Diffusion-reaction equations were originally developed to model chemical reactions involving microscopic particles in a fluid undergoing random, Brownian motion. The diffusion-reaction equation is given by,

$$\frac{\partial u}{\partial t} = D\nabla^2 u + R, \tag{1}$$

where $u = u(x, y, t)$ is some unknown function, D is a matrix of diffusion coefficients, modelling the net movement of particles from an area of high concentration to an area of low concentration, and $R = R(u, x, y, t)$ is the reaction term, which models how particles change from one type to another. This equation is ideal for modelling disease spread since, on a large scale, the motion of people can be assumed to be random.

In this model, we define,

$$u(x, y, t) = S(x, y, t) + I(x, y, t) + R(x, y, t), \tag{2}$$

as the total population, which must be conserved. We have defined the following additional fields:

- $S(x, y, t)$ is the susceptible population (those who can catch the disease).
- $I(x, y, t)$ is the infected population (those who have the disease and can spread it).
- $R(x, y, t)$ is the recovered population.

We then define a reaction term consisting of two parts, the rate of new infection,

$$\frac{\partial S}{\partial t} = -\beta_0 \rho(x, y) SI, \tag{3}$$

and the rate of recovery,

$$\frac{\partial R}{\partial t} = \gamma I, \tag{4}$$

where β is the rate of infection, $\rho(x, y)$ is the population density field, and

$$\gamma = \frac{1}{\tau}, \tag{5}$$

where τ is the recovery time. This allows us to obtain the final differential equation,

$$\frac{\partial I}{\partial t} = D\nabla^2 u(x, y, t) + \beta_0 \rho(x, y) SI - \gamma I. \tag{6}$$

We chose to model COVID-19, since a large amount of data on its spread is available to compare the model to, as well as its high rate of infection.

Theory - Numerical Methods

The system of equations was discretised using the method of lines and a five-point stencil, meaning each spatial cell has four nearest-neighbours that contribute to its spatial gradient, as below,

$$\nabla^2 I \approx I_{i+1,j} + I_{i-1,j} + I_{i,j+1} + I_{i,j-1} - 4I_{i,j}. \tag{7}$$

Each borough within Greater Manchester was modelled using a Gaussian distribution centred on the area, using census data to obtain population densities.

Results: Model Figure

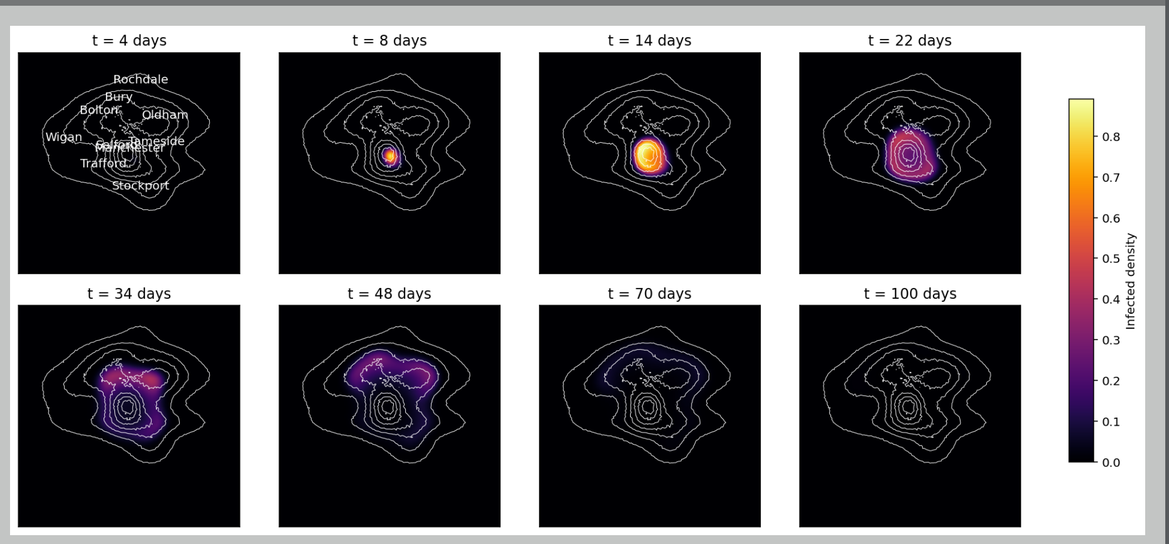

Figure 1: COVID-19 infection over time in Greater Manchester up to 100 days. Using parameters of initial transmission rate $\beta_0 = 0.48$ per day and diffusion coefficient of $D = 0.85$ m^2 s^{-1} and a mean recovery time of $\tau = 7$ days.

Results: Simulation vs Data

Infection of COVID-19 was modelled in Greater Manchester with an initial transmissibility of $\beta_0 = 0.48$ per day taking into account weather effects [2] while using a diffusion coefficient of $D = 0.85$ m^2 s^{-1}, and a mean recovery time of $\tau = 7$ days [3] and using the spatial density given for each county of Greater Manchester from the March 2021 Census data [4]. The model was compared to the COVID-19 infection map of Boston [5] as it is a city of comparable size thus allowing us to clearly compare the accuracy of our model. As seen our model is comparable to the Boston infection map up to the first 30 days. After that, the model diverges and while spreading out the infection at the initial points in the centre of Greater Manchester disappears as opposed to the Boston heat map.

Results: Spread of Covid From Data

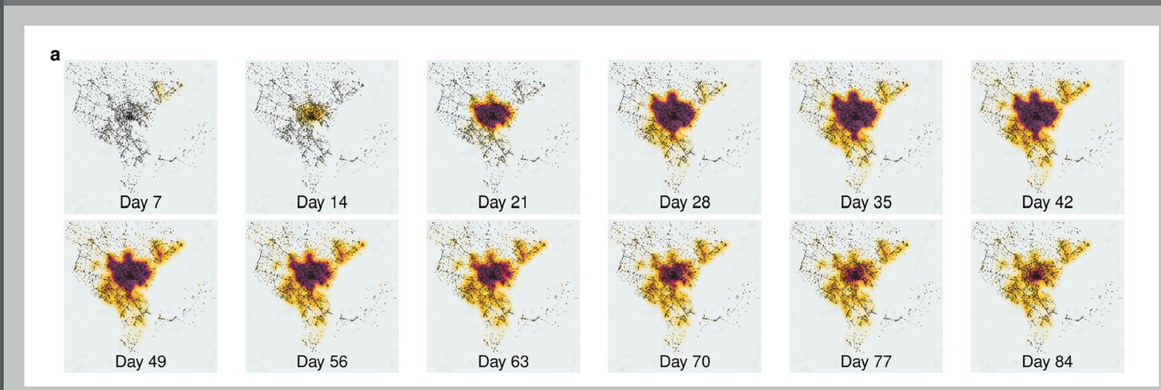

Figure 2: Covid-19 infection "Heatmap of infected (exposed and infectious) individuals every seven days: (a) Auto Sprawl;"[5] of Boston over 84 days. Figure 5 from " Activity-based epidemic propagation and contact network scaling in auto-dependent metropolitan areas" by Kumar, Nishant & Oke, Jimi & Nahmias Biran, Bat-hen,2021, Scientific Reports, Volume 11, p. 9. Copyright 2021 by Authors. CC BY "

Discussion

We found that though our model was successful at modelling the initial spread of the Covid 19 pandemic. In particular, the initial growth seen in Fig.1 matches near perfectly data seen in Fig. 2. However, after this, the model starts to diverge from real life data. This is likely because we made the assumption that people's behaviour wouldn't change when infected and the nature of the Reaction-Diffusion equation used in the model. In reality, as time went on, and more was understood about Covid, people started to self isolate when they first experienced symptoms, and the first lockdowns started to occur in March 2020 [6]. However, this doesn't make the idea of using diffusion-reaction useless or invalid in all cases. Instead, it implies that the best use of such a model is to predict what would happen should the spread of the virus not be contained for the initial spread.

Our model also fails to account for individuals travelling long distances, instead assuming that everyone remains in roughly the same location. In reality, this isn't true for multiple reasons. Firstly, as a big city, people commute into central Manchester from the likes of Salford, Stockport, and Tameside both to shop and work. These people are likely to get infected in Manchester and travel back to their homes, causing the pandemic to spread to outer regions sooner than in the model. Greater Manchester isn't closed either with both national and international travel happening in and out daily. Whilst an infected person leaving Manchester wouldn't change much, at least outside of the very early days, an infected person entering Manchester could lead to a new infection centre. In reality, it turns out that, "connectivity matters more than density in the spread of COVID-19" [7].

A diffusion-reaction model also fails to account for the virus, "Sticking around", as seen in Fig. 2. In our model, after around 30 days the area where our "patient zero" first appeared had been near completely cleared out as everyone in the area had caught the illness and recovered. This contrasts with the data seen above. In reality, not everyone in an area gets ill immediately, so the illness will stay around for longer rather than just spread to other areas. As such, even months on the infected density for the virus doesn't fall off close to 0 unlike our model. Despite this, the model may still be valid. Historically pandemics tend to slow down as they burn through all people in an area as herd immunity protects the population. As real life lockdowns (which aren't taken into account by our model) slowed down the spread of infection, it took far longer for herd immunity to be achieved in real life.

Conclusions

In conclusion, the reaction-diffusion equation successfully captured the initial spread of COVID-19, but diverged from real-world data at later times. This is due to simplifying assumptions, such as a lack of behavioural changes, a travelling population, and the virus not lingering in certain areas, which a better model for disease spread would account for.

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